

Quiz — 1.2.3 Software Development

Name: _____ Date: _____ Mark: _____ / 25

OCR H446 · Component 01 · 1.2.3

Section A — Multiple choice (8 marks)

1. Which methodology is **linear/sequential**, completing one stage fully before the next begins? A. Agile B. Waterfall C. Spiral D. RAD Your answer: ☐
 2. A defining feature of the **Agile** approach is: A. A single delivery of the finished product at the end B. Iterative increments with continuous client feedback C. Heavy upfront documentation with no later changes D. Risk analysis at the centre of every loop Your answer: ☐
 3. **Pair programming** and test-driven development are most associated with: A. Waterfall B. Extreme programming (XP) C. Spiral D. RAD Your answer: ☐
 4. The **Spiral** model is best suited to projects that are: A. Small with fixed, clear requirements B. Large, expensive and high-risk C. Needed extremely quickly with vague requirements D. Trivial throwaway scripts Your answer: ☐
 5. **RAD** primarily relies on: A. Detailed sequential documentation B. Rapid prototyping refined by user feedback C. A central risk-analysis loop D. One fixed requirements specification Your answer: ☐
 6. During which lifecycle stage are **requirements gathered** and **feasibility** assessed? A. Design B. Analysis C. Implementation D. Evaluation Your answer: ☐
 7. Testing a program with **boundary data** means using values that are: A. Far outside the acceptable range B. At the edges of the acceptable range C. Always invalid/erroneous D. Random and unrepeatable Your answer: ☐
 8. The **Evaluation** stage judges the finished system mainly against: A. The personal preferences of the lead programmer B. The original requirements (effective, usable, robust, maintainable) C. The number of lines of code written D. The cost of the hardware Your answer: ☐
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Section B — Short answer (12 marks)

9. State **two** advantages and **one** disadvantage of the Waterfall methodology. [3]

10. Explain why Agile is often more suitable than Waterfall when a client's requirements are **unclear or likely to change**. [2]

11. Name and describe the three categories of test data used to test a program thoroughly. [3]

12. Describe **two** tasks carried out during the **Design** stage of the software lifecycle. [2]

13. Explain what is meant by **risk analysis** in the Spiral model and why it benefits a high-cost project. [2]

Section C — Exam-style question (5 marks)

14. A startup is building a brand-new mobile app. The client is unsure exactly what features they want and expects ideas to change as early versions are shown to real users. The budget is modest and the team is small but skilled. Recommend a **suitable development methodology** and justify your choice with reference to the scenario. [5]

[illegible]